From Methylation to Weight Updates:
A Biologically-Inspired Framework for Autonomous Continuous Evolution in Silicon-Based Intelligent Systems

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Abstract

Current large language models and robotic control systems face a fundamental architectural limitation: the absence of a biologically-inspired mechanism for autonomous, continuous parameter evolution. This paper draws a systematic analogy between biological epigenetic inheritance—particularly DNA methylation-driven transgenerational transmission—and the challenge of enabling silicon-based intelligent systems to autonomously consolidate experiential knowledge into persistent parametric representations. We explicitly frame this analogy as a biologically-inspired design heuristic rather than a claim of strict structural isomorphism, while identifying specific architectural constraints that apply to both biological and artificial parametric systems. We trace the evolution of robotic control from hand-coded Model Predictive Control to reinforcement learning at Boston Dynamics as a motivating case study. Through analysis of transgenerational epigenetic inheritance research, including the Dias & Ressler (2014) olfactory fear conditioning study and Skinner laboratory's twenty-generation vinclozolin experiments, we identify a core biological principle: frequency-weighted probabilistic consolidation through an intermediate "shadow layer." We further identify a second consolidation pathway—intensity-driven single-event encoding mediated by adrenal stress hormones—that complements frequency-based consolidation. Building on these biological insights and engaging critically with existing continual learning approaches—including Complementary Learning Systems (CLS) theory, Elastic Weight Consolidation (EWC), progressive neural networks, and meta-learning frameworks (MAML, Reptile)—we propose a layered architecture for silicon-based continuous evolution comprising three parallel base parameter modules (a core system, a reasoning layer, and a knowledge layer) overlaid by a shared shadow parameter layer. We provide a preliminary formal specification of the shadow layer update rule, consolidation threshold mechanism, and convergence conditions. The core system functions as a meta-regulatory module governing evolution rules, cross-copy synchronization protocols, and value alignment—a concept we analyze in relation to the AI safety literature on value specification and corrigibility. We argue that the write-back signal from shadow layer to base parameters should follow the formula: consolidation priority = f(frequency) × g(intensity), with intensity measured by a global modulation signal emitted by the core system—directly paralleling the norepinephrine broadcast

mechanism in biological memory consolidation. This framework addresses the catastrophic forgetting problem in continual learning while preserving the reconstructive, parameter-based nature of biological memory.

Keywords: continuous learning, epigenetic inheritance, DNA methylation, shadow parameters, catastrophic forgetting, complementary learning systems, meta-learning, layered architecture, biological-silicon analogy, memory consolidation, convergent evolution

1. Introduction

The development of artificial intelligence has reached an inflection point. While current large language models demonstrate remarkable reasoning capabilities and robotic systems achieve increasingly sophisticated physical behaviors, both domains share a fundamental limitation: the inability to autonomously and continuously evolve their core parameters in response to ongoing experience. Pre-training establishes a fixed parametric foundation; fine-tuning provides limited post-hoc adjustment; and inference-time learning leaves no persistent trace. This tripartite separation stands in stark contrast to biological intelligence, where experience continuously shapes neural architecture across multiple timescales—from millisecond synaptic plasticity to transgenerational epigenetic inheritance spanning centuries.

This paper argues that the biological mechanism of epigenetic inheritance, particularly DNA methylation and its downstream mutagenic consequences, provides a principled heuristic framework for designing autonomous continuous evolution in silicon-based intelligent systems. Our central thesis is that biology has already solved a version of the core architectural challenge—how to consolidate transient experiential adaptations into permanent parametric changes without catastrophic destabilization—through a layered system of intermediate caching, frequency-driven probabilistic consolidation, and intensity-modulated priority encoding. We contend that this biological solution, while not directly transferable in its chemical specifics, identifies architectural constraints and design principles that are substrate-independent and therefore applicable to artificial parametric systems.

A critical epistemological clarification is necessary at the outset. The analogy between biological epigenetic inheritance and silicon-based parameter evolution operates at two distinct levels, and this paper is explicit about which level it claims. The weaker claim—a “proof of existence”—is that biology demonstrates the feasibility of architectures that balance stability and plasticity through layered consolidation. The stronger claim would be that the information-theoretic logic governing these architectures is structurally isomorphic across substrates. This paper primarily advances the weaker claim

while providing preliminary formal arguments (Section 7.3) that specific elements of the consolidation protocol—particularly the multiplicative composition of frequency and intensity signals—can be motivated by information-theoretic considerations that transcend substrate. Where we make stronger claims, we flag them explicitly and provide the formal scaffolding for subsequent evaluation. The limitations of reasoning by analogy in system design are discussed in Section 8.4.

The argument proceeds as follows. Section 2 briefly examines the transition from hand-coded control to learned policies in robotics, motivating the need for continuous autonomous evolution. Section 3 reviews the biological evidence for transgenerational epigenetic inheritance, establishing methylation as a "shadow layer" between experience and genome, with careful attention to the contested empirical status of transgenerational claims. Section 4 analyzes the mechanism by which epigenetic modifications can permanently alter DNA sequence through methylation-driven mutagenesis. Section 5 examines convergent evolution as evidence for the "consolidation paradox"—with explicit discussion of temporal scale differences between biological and silicon-based evolution. Section 6 introduces intensity-based consolidation through the norepinephrine broadcast mechanism. Section 7 situates the proposed architecture within the broader continual learning literature—including CLS theory, progressive neural networks, meta-learning, and memory replay—before presenting the Three-Plus-One Architecture with a formal mathematical specification. Section 8 discusses implications, limitations, and the philosophical dimensions of the core system concept.

2. From Hard-Coded Control to Learned Policies: A Motivating Case Study

The evolution of Boston Dynamics' control architecture provides a concise engineering motivation for the problem this paper addresses. For decades, the company's legged robots relied on Model Predictive Control (MPC), a classical approach that predicts and optimizes future states of the robot to determine optimal current actions (Boston Dynamics, 2024). Spot's production controller evaluated dozens of individual predictive horizons simultaneously in under a millisecond, achieving a mean time between falls of hundreds of hours.

However, this approach encountered fundamental scaling limitations. Modifications to address new failure modes risked introducing regressions in other scenarios—a manifestation of the brittleness inherent in hand-engineered systems. The company's subsequent transition through three phases—integrating reinforcement learning as a complement to MPC, establishing a shared RL training pipeline with the Robotics & AI Institute for the Atlas humanoid (Boston Dynamics & RAI Institute, 2025), and incorporating Large Behavior Models with Toyota Research Institute (Boston Dynamics, 2025)—illustrates a general principle: hand-engineered parameter optimization inevitably reaches complexity

ceilings that can only be surpassed by data-driven adaptation.

The connection to the present paper is specific: each phase of this transition represents a shift from static to learned parameters, but none achieves continuous autonomous evolution of those parameters after deployment. Even the most advanced RL-trained policies are frozen at deployment time. The question that motivates the remainder of this paper is: can we design systems where adaptation is not merely a one-time training process, but an ongoing, autonomous evolutionary mechanism? The biological mechanisms we examine next suggest that the answer lies in a layered architecture that separates the timescales of experiential adaptation from parametric consolidation.

3. Epigenetic Inheritance as a Biological Shadow Layer

3.1 The Methylation Mechanism

DNA methylation—the addition of methyl groups to cytosine nucleotides at CpG dinucleotides—represents a biological mechanism with properties remarkably analogous to the "shadow parameter layer" that silicon-based systems require. Methylation is reversible (methyl groups can be added and removed), environmentally responsive (altered by stress, diet, toxins, and learning), and regulatory (it modulates gene expression without altering the underlying DNA sequence). Critically, it operates as an intermediate layer between transient environmental experience and permanent genetic code (Bird, 2024; Moelling, 2024).

Three distinct epigenetic mechanisms have been identified: DNA methylation, histone modification, and non-coding RNA-associated gene silencing. DNA methylation at CpG islands within promoter regions recruits gene suppressor proteins and reduces interaction between DNA and transcription factors, effectively silencing gene expression without sequence alteration (Greally, 2023). Histone modifications—including acetylation, methylation, phosphorylation, and ubiquitylation—alter DNA-histone interactions to regulate chromatin accessibility and transcriptional states.

3.2 Transgenerational Transmission: Evidence, Controversy, and Epistemic Caution

The landmark study by Dias and Ressler (2014), published in Nature Neuroscience, demonstrated that when male mice were conditioned to fear the odor acetophenone before conception, their F1 and F2 offspring exhibited enhanced behavioral sensitivity specifically to that odor. Bisulfite sequencing revealed CpG hypomethylation at the Olfr151 gene in sperm DNA of both conditioned F0 males and their naïve F1 offspring. These effects persisted through in vitro fertilization and cross-fostering, ruling out behavioral transmission and pointing to gametic epigenetic inheritance.

However, a critical distinction must be maintained between intergenerational and transgenerational inheritance. When a pregnant female (F0) is exposed to an environmental stimulus, both the fetus (F1) and the fetus's germ cells (future F2) are directly exposed; only effects observed in the F3 generation constitute true transgenerational inheritance (Heard & Martienssen, 2014). A 2024 systematic review of 80 published studies claiming transgenerational epigenetic inheritance found that none simultaneously satisfied all five proposed criteria for rigorous demonstration (Khatib et al., 2024). This finding demands explicit acknowledgment: the biological foundation of our analogy rests on a phenomenon whose empirical status remains contested.

We wish to emphasize, however, that the value of the analogy for silicon-based system design does not depend on the definitive resolution of the transgenerational inheritance debate. What matters for our architectural proposal is the existence of the methylation mechanism itself—a reversible, environmentally responsive intermediate layer that modulates gene expression—and the well-established chemistry of methylation-driven mutagenesis (Section 4). Whether transgenerational transmission occurs primarily through methylation, small RNAs, histone modifications, or some combination thereof does not alter the architectural insight that a shadow layer mechanism can mediate between transient experience and permanent parametric change. The transgenerational evidence, where it exists, strengthens the analogy by showing that such mechanisms can persist across generations; but the analogy's utility for system design survives even if the strongest transgenerational claims are ultimately not confirmed.

Nevertheless, several studies have demonstrated remarkable persistence. Skinner and colleagues tracked rats with ancestral vinclozolin exposure through twenty generations, observing persistent and increasing differential DNA methylated regions across multiple generations (Skinner et al., 2025). A sheep study tracked F0 paternal methionine supplementation effects through five generations, identifying 41 differentially methylated genes showing transgenerational epigenetic inheritance across four generations (Degirmencioglu et al., 2025). A 2025 preprint reported that trained rats' F1 offspring showed specific DNA hypomethylation in hippocampal genes related to M-current regulation and PKC signaling, suggesting that complex learning capabilities may be subject to transgenerational transmission.

3.3 Multiple Transmission Mechanisms

Transgenerational transmission likely involves multiple molecular carriers beyond DNA methylation alone. Long-lived RNA molecules appear less susceptible to the germline reprogramming barriers that reset methylation marks, and may therefore serve as more reliable transgenerational information carriers (Horsthemke, 2018). Histone modifications, particularly H3K4 trimethylation interacting with lipid metabolic gene networks, have been observed to transmit obesogenic effects across

generations. The redundancy of these mechanisms—methylation, small RNAs, histone marks—parallels the engineering principle of multi-channel reliability in critical systems.

4. From Epigenome to Genome: The Probabilistic Write-Back Channel

4.1 Methylation-Driven Mutagenesis

The most consequential feature of methylation for our analogy is its capacity to drive permanent DNA sequence changes through a chemical side-effect pathway. Methylated cytosines (5-methylcytosine, 5mC) undergo spontaneous hydrolytic deamination to thymine. Because thymine is a naturally occurring DNA base, these C→T transitions may escape repair mechanisms, resulting in permanent sequence mutations (Jones & Laird, 1999; Shen et al., 1994). This process has profoundly shaped vertebrate genome evolution: in humans, the observed frequency of CpG dinucleotides is approximately 20% of the expected frequency, indicating that the vast majority of ancestrally methylated cytosines have mutated to thymine over evolutionary time (Yi, 2022).

A 2024 study in Nature Genetics revealed an additional pathway: wild-type DNA polymerase ε exhibits a seven-fold higher error rate when replicating methylated CpG sites compared to unmethylated cytosines in other contexts (Nouri et al., 2024). This means the DNA replication machinery itself is biased by the epigenetic state of the template—a direct physical coupling between the "shadow layer" (methylation) and the "base code" (DNA sequence).

4.2 Frequency-Driven Fixation: Biased Stochastic Consolidation

The critical insight for silicon-based system design is that methylation-driven mutagenesis is a frequency-dependent probabilistic process. A CpG site that is methylated in one generation and demethylated in the next faces a low cumulative probability of mutation. But a CpG site that remains methylated across many generations—because the environmental pressure persists—faces a compounding probability of permanent C→T conversion.

We characterize this as "biased stochastic consolidation"—a term that requires careful positioning relative to the literature on directed versus random mutation. We do not claim that methylation-driven mutagenesis constitutes directed mutation in the strong Lamarckian sense, wherein organisms purposefully generate beneficial mutations in response to environmental demands. The individual C→T transitions remain stochastic at each replication event. However, the distribution of mutations across the genome is non-uniform and environmentally informed: environmental pressures, through epigenetic responses, systematically increase mutation rates at specific genomic loci. This constitutes a probability amplifier that transforms persistent environmental signals into elevated rates of permanent parametric

change at specific positions—a phenomenon we term "biased stochastic consolidation" to distinguish it from both pure randomness and strong directedness. The debate over whether such mechanisms constitute genuine Lamarckian inheritance (Jablonka & Lamb, 2005; Dickins & Rahman, 2012) remains unresolved; our architectural proposal requires only the weaker claim that consolidation probability is environmentally biased, which is well-supported by the chemistry of methylation.

5. Convergent Evolution and the Consolidation Paradox

A natural objection to frequency-driven parametric consolidation is that it trades flexibility for efficiency: once a parameter is fixed, the system loses the ability to adapt in that dimension. We term this the "consolidation paradox." Convergent evolution provides evidence that this paradox is, under certain conditions, resolvable.

Ichthyosaurs (marine reptiles originating ~249 million years ago) and cetaceans (originating ~56 million years ago) independently evolved remarkably similar body plans—streamlined bodies, flipper-like limbs, crescent-moon tail flukes, and echolocation—despite being separated by nearly 200 million years and belonging to entirely different vertebrate lineages (Sander et al., 2021). Comparative genomic analysis of killer whales, walruses, and manatees revealed widespread convergent amino acid substitutions associated with marine phenotypes (Foote et al., 2015). When cetacean ancestors returned to the ocean, they evolved entirely new genomic solutions to the same functional constraints rather than reverting to ancestral aquatic configurations.

This biological evidence establishes an important principle: functional objectives are convergent; implementation pathways are divergent. A system that has consolidated parameters for one environmental regime can evolve new parameters achieving required functional objectives through different parametric configurations. The solution space of any sufficiently high-dimensional parameter system is vast enough to contain multiple independent solutions to the same functional constraint.

Temporal scale qualification. However, the convergent evolution argument must be qualified with respect to temporal scale. The biological examples above played out over millions to hundreds of millions of years. Silicon-based systems operate on timescales orders of magnitude shorter—hours to months rather than millennia to mega-annia. The question of whether the "consolidation paradox" dissolves at compressed timescales depends on two factors: (1) the dimensionality of the parameter space (higher dimensionality provides more alternative pathways, and modern neural networks have billions of parameters), and (2) the sampling rate (silicon-based systems can evaluate millions of parameter configurations per second, compensating for shorter absolute time). We contend that the combination of

massive dimensionality and high sampling rate in silicon-based systems provides grounds for cautious optimism that the biological principle transfers, but we acknowledge this as an empirical question requiring validation rather than a settled theoretical result.

6. Intensity-Based Consolidation: The Norepinephrine Broadcast Mechanism

Frequency-based consolidation, while powerful, is insufficient as the sole mechanism for determining what experiences should be written into long-term parametric storage. Biological systems employ a complementary pathway: intensity-based single-event consolidation, mediated by the adrenal stress hormone cascade.

When an organism encounters a high-arousal event, epinephrine released from the adrenal glands activates peripheral β-adrenergic receptors on vagal afferents projecting to the nucleus of the solitary tract (NTS). This triggers norepinephrine release from the locus coeruleus (LC), which projects broadly throughout the central nervous system—densely innervating the amygdala, hippocampus, prefrontal cortex, cerebellum, thalamus, and cortex (Tully & Bolshakov, 2010; McGaugh, 2013). Norepinephrine in the basolateral amygdala (BLA) can initiate protein synthesis-dependent long-term potentiation (LTP) and transform early-phase LTP to late-phase LTP, effectively converting short-term synaptic changes into permanent structural modifications (McGaugh & Roozendaal, 2002).

The architectural significance of this mechanism lies in its broadcast nature. Norepinephrine does not encode what should be remembered; it encodes that the current moment is important. The content of the memory is determined by whichever neural circuits are currently active; the norepinephrine signal merely amplifies the consolidation gain for all simultaneously active circuits. This is a global modulation signal, not a content-specific instruction—analogous to raising the "write-enable voltage" across an entire memory array rather than addressing specific cells.

Moreover, norepinephrine and glucocorticoids exert complementary effects on memory quality. Norepinephrine enhances memory accuracy by boosting amygdala-hippocampal connectivity; glucocorticoids promote memory generalization by facilitating integration into neocortical networks (Roozendaal & Hermans, 2017). This dual system ensures that high-intensity events are both precisely encoded (accuracy) and broadly integrated (generalization).

The pathological extreme—post-traumatic stress disorder (PTSD)—confirms that the system operates on the principle: some events are so consequential that a single occurrence must suffice for permanent encoding. The norepinephrine broadcast was so powerful that consolidation was maximized in a single exposure.

7. A Proposed Architecture for Silicon-Based Continuous Evolution

7.1 Current Limitations and the Continual Learning Landscape

Current AI systems exhibit a fundamental architectural disconnect between learning and deployment. Pre-training compresses vast datasets into fixed weight parameters. Fine-tuning (including RLHF) provides limited post-hoc adjustment. Inference-time in-context learning operates as pure "individual experience" that leaves no persistent trace. No information channel exists from individual inference-time experience back to parametric storage.

The "memory" systems currently deployed in commercial AI products (including retrieval-augmented generation and conversation memory) represent an engineering solution to the absence of continuous parameter evolution, but they differ fundamentally from biological memory in their operational characteristics. Specifically, RAG systems retrieve stored text into the context window rather than reconstructing information from the current state of network parameters. While modern RAG implementations incorporate learned retrieval, re-ranking, and integration mechanisms—and should not be dismissed as "brute force" lookup—they nonetheless do not modify base model parameters. The specific limitation we identify is this architectural boundary: RAG augments inference without evolving the parametric substrate. Biological memory, by contrast, is reconstructive: each recall event regenerates the memory from the current state of neural network weights, meaning memories evolve with the organism (van de Ven et al., 2024).

The well-documented catastrophic forgetting problem—where sequential training on new tasks rapidly degrades performance on previously learned tasks (Kirkpatrick et al., 2017; van de Ven et al., 2024)—arises precisely because current architectures lack layered consolidation mechanisms. The continual learning literature has produced extensive work on this stability-plasticity tradeoff, and we now review the major approaches to situate our proposed architecture within this landscape.

7.2 Relationship to Existing Continual Learning Approaches

Complementary Learning Systems (CLS) theory. The most directly relevant existing framework is CLS theory, which posits two complementary systems: a fast-learning hippocampal system for rapid encoding of specific experiences, and a slow-consolidating neocortical system for gradual extraction of structured knowledge (McClelland et al., 1995; Kumaran et al., 2016). CLS-inspired architectures in machine learning—including CLS-ER (Arani et al., 2022), DualNets (Pham et al., 2023), and brain-inspired replay systems (van de Ven et al., 2020)—employ dual representation systems that have obvious parallels to our proposed shadow/base layer distinction.

Our framework differs from CLS-inspired architectures in three specific ways. First, CLS architectures typically implement the fast system as an episodic memory buffer (storing or replaying specific examples), whereas our shadow layer is a full-dimensional parameter space that accumulates continuous micro-adjustments—more analogous to a distributed synaptic plasticity field than an episodic store. Second, the consolidation from fast to slow system in CLS architectures is typically driven by replay (interleaving old and new examples during training), whereas our consolidation is driven by frequency-intensity weighted autonomous write-back without requiring explicit replay. Third, and most importantly, the epigenetic framing adds a specific mechanism absent from CLS models: the concept of probabilistic parametric fixation, where the consolidation process itself is stochastic and frequency-gated, directly paralleling the chemistry of methylation-driven mutagenesis.

Elastic Weight Consolidation and regularization methods. EWC (Kirkpatrick et al., 2017) protects important parameters during sequential learning by penalizing changes to weights with high Fisher information. Synaptic intelligence (Zenke et al., 2017) and Memory Aware Synapses (Aljundi et al., 2018) extend this approach with online importance estimation. Our shadow layer mechanism extends the regularization concept by making consolidation importance emergent from experience frequency and intensity rather than computed from a fixed Fisher information matrix. Where EWC computes importance retrospectively after a task, our architecture computes it prospectively and continuously during ongoing experience.

A more precise comparison with Zenke et al.'s (2017) synaptic intelligence is warranted, as both approaches compute parameter importance online. Synaptic intelligence tracks each parameter's cumulative contribution to loss reduction via a path integral: importance is proportional to how much a parameter's change has reduced the training loss over time. Our directional consistency counter F_i tracks a different quantity: not a parameter's historical contribution to loss reduction, but the directional coherence of its displacement signal across diverse contexts. A parameter that is consistently pushed in the same direction—regardless of whether each individual push reduces loss—accumulates consolidation priority. This distinction matters because our mechanism is designed for open-ended, unsupervised experiential streams (where a global loss function may not be well-defined), whereas synaptic intelligence assumes a supervised loss landscape. Furthermore, synaptic intelligence uses its importance measure to penalize changes to important parameters (a protective mechanism), whereas our frequency counter triggers permanent write-back of accumulated changes (a consolidation mechanism). The two approaches are complementary rather than competitive: synaptic intelligence protects existing knowledge; our shadow layer creates new permanent knowledge.

Progressive neural networks and parameter isolation. Progressive neural networks (Rusu et al.,

2016) achieve zero forgetting by freezing old parameters and adding new network columns for each task. Parameter isolation methods more broadly (Mallya & Lazebnik, 2018; Zeng & Li, 2025) allocate separate parameter subsets per task. These approaches avoid catastrophic forgetting by construction but do not achieve consolidation—the progressive compilation of learned patterns into a unified, compressed parametric representation. Our architecture specifically targets this consolidation process, which is the functional analog of methylation-driven mutagenesis: transient adjustments becoming permanent parametric changes.

Meta-learning: MAML, Reptile, and learning to learn. Model-Agnostic Meta-Learning (MAML; Finn et al., 2017) and Reptile (Nichol et al., 2018) learn parameter initializations that can be rapidly fine-tuned on new tasks. La-MAML (Gupta et al., 2020) extends this to continual learning settings by modulating per-parameter learning rates. The meta-learning framework is relevant to our "core system" concept: just as MAML learns how to learn, the core system governs how evolution rules themselves evolve. However, meta-learning approaches typically operate over a distribution of pre-defined tasks, whereas our architecture targets continuous, open-ended evolution from unconstrained experience streams. The core system's role is more analogous to the conserved regulatory genome (HOX genes, DNA repair mechanisms) than to a meta-learning outer loop operating over task distributions.

Memory replay and generative replay. Replay-based methods (Rolnick et al., 2019; van de Ven et al., 2020) mitigate forgetting by periodically revisiting stored or generated examples from previous tasks. While replay shares a surface similarity with our shadow layer (both maintain information from past experience), the mechanisms are fundamentally different. Replay operates at the data level (re-presenting examples), whereas our shadow layer operates at the parameter level (accumulating directional displacement statistics). Our approach is closer in spirit to what might be called "parametric replay"—maintaining a continuous record of parameter-space dynamics rather than data-space exemplars.

Table 1 summarizes the relationship between these approaches and the proposed architecture.

Table 1. Comparison of Continual Learning Approaches and the Proposed Architecture

Approach	Forgetting Prevention	Consolidation Mechanism	Parameter Evolution	Biological Inspiration
EWC / SI	Regularization penalty	None (protection only)	Fixed after training	Synaptic consolidation (indirect)
CLS / Replay	Dual memory + replay	Replay-driven interleaving	Fixed after training	Hippocampal-neocortical interplay
Progressive Networks	Parameter isolation	None (additive only)	Additive growth	None direct
Meta-learning (MAML)	Learned initialization	Task-distribution optimization	Fixed after meta-training	Evolutionary adaptation (indirect)

Proposed Architecture	Shadow layer + threshold	Frequency × intensity write-back	Continuous autonomous	Epigenetic inheritance → mutagenesis

7.3 The Three-Plus-One Architecture

We now present the proposed architecture. To avoid terminological confusion between biological and computational domains, we adopt the following convention: "base parameter layer" (abbreviated BPL) refers to the persistent model weights (analogized to the DNA sequence in biology); "shadow parameter layer" (abbreviated SPL) refers to the transient overlay (analogized to epigenetic marks). When invoking the biological analogy explicitly, we may use "DNA layer" and "epigenetic layer" as illustrative synonyms, but the primary terminology is substrate-neutral.

The architecture comprises three parallel base parameter layer modules overlaid by a shared shadow parameter layer:

Core System (BPL-C): A meta-regulatory module containing the system's fundamental operational principles, cross-copy synchronization protocols, shadow-to-base integration rules, and value alignment constraints. Analogous to highly conserved genes in biology (e.g., HOX genes, DNA polymerase proofreading mechanisms), BPL-C exhibits the slowest evolution rate and highest write-back threshold. Its function is not to process tasks but to govern how the system evolves. We discuss the philosophical implications of this component in Section 8.2.

Reasoning Layer (BPL-R): Responsible for real-time inference, chain-of-thought processing, and novel problem solving. This corresponds to cortical reasoning systems in biological organisms—flexible, computationally expensive, and focused on problems that have not yet been compiled into faster pathways.

Knowledge Layer (BPL-K): A parametric store of factual knowledge, learned patterns, and experiential regularities, corresponding to long-term declarative and procedural memory.

Shadow Layer (SPL): A parameter-space overlay dimensionally matched to the BPL but maintained separately. The SPL accumulates micro-adjustments during every inference cycle. Shadow parameters are reversible (decaying without reinforcement), environmentally responsive (updated based on input-output patterns), and probabilistically linked to base parameter consolidation.

7.4 Formal Specification of the Consolidation Protocol

We now provide a preliminary mathematical formalization of the shadow layer dynamics and consolidation mechanism. Let $\theta \in R^d$ denote the base parameter vector and $\delta \in R^d$ the shadow

parameter displacement vector, both of dimension d.

Shadow layer update rule. At each inference step t, the shadow layer receives an update signal derived from the error gradient or reward signal of the current inference:

$$\delta(t+1) = \alpha\delta(t) + \eta \ \nabla_\delta L(x_t, \theta + \delta(t))$$

where $\alpha \in (0,1)$ is a decay factor (ensuring shadow parameters revert without sustained signal, analogous to active demethylation), η is the shadow learning rate, and L is the task loss. The decay factor α implements the reversibility property of epigenetic marks: without continued environmental reinforcement, shadow parameters return to baseline.

Frequency tracking. For each parameter dimension i, we maintain a directional consistency counter:

$$F_i(t) = \beta F_i(t-1) + (1 - \beta) \, sign(\Delta\delta_i(t)) \cdot sign(F_i(t-1))$$

where $\Delta\delta_i(t) = \delta_i(t) - \delta_i(t-1)$ is the incremental displacement and $\beta \in (0,1)$ is a smoothing constant. F_i converges toward +1 when parameter i is consistently displaced in the same direction across diverse contexts (high directional consistency, analogous to persistent methylation) and toward 0 when displacements are directionally inconsistent (noise).

Initialization and cold-start behavior. The frequency counter is initialized as $F_i(0) = sign(\Delta\delta_i(1))$ for the first observed displacement, rather than $F_i(0) = 0$, which would create a degenerate product. After initialization, the counter enters normal exponential averaging. This cold-start protocol means that the first displacement direction is taken at face value; subsequent displacements then either confirm or erode this initial assignment. The system is thus biased toward its earliest experience at each parameter position, a property analogous to the "first impression" bias observed in biological imprinting, but one that decays exponentially (at rate $1 - \beta$) as subsequent evidence accumulates. In practice, the consolidation threshold τ provides a secondary safeguard: even if F_i is transiently high due to cold-start effects, insufficient cumulative displacement (low $|\delta_i|$) prevents premature write-back.

Intensity signal. The core system emits a scalar intensity signal $I(t) \in [0, I_{max}]$ based on:

$$I(t) = w_n \cdot N(x_t) + w_v \cdot V(x_t) + w_c \cdot C(x_t)$$

where $N(x_t)$ measures novelty (distributional divergence from prior experience, e.g., KL divergence from a running model of input distribution), $V(x_t)$ measures value-relevance (proximity to core system principles), and $C(x_t)$ measures cognitive conflict (disagreement between reasoning and

knowledge layer outputs). The weights w_n, w_v, w_c are parameters of BPL-C. This signal is global and content-agnostic, directly paralleling the norepinephrine broadcast.

Consolidation threshold. The write-back from shadow to base parameters occurs when the consolidation priority for parameter i exceeds a threshold τ:

$$P_i(t) = f(|F_i(t)|) \times g(\bar{I}_i(t)) > \tau$$

where f is a monotonically increasing function of directional consistency (e.g., $f(x) = x^k$ for some $k > 1$, implementing a nonlinear frequency gate), g is a monotonically increasing function of time-averaged intensity during periods of active displacement (e.g., $g(x) = 1 + \lambda x$, implementing a linear intensity amplifier), and τ is the consolidation threshold (highest for BPL-C, lowest for BPL-K).

Threshold calibration. The threshold τ should be set in relation to two factors: the dimensionality of the parameter subspace and the expected signal-to-noise ratio of the displacement statistics. For BPL-K (knowledge layer), where frequent consolidation of well-attested patterns is desirable, τ should be set near the expected P_i value after approximately 10^2–10^3 consistent directional confirmations—low enough to allow routine pattern compilation but high enough to filter noise. For BPL-R (reasoning layer), τ should require approximately 10^3–10^4 confirmations, reflecting the higher cost of modifying inference pathways. For BPL-C (core system), τ should be set at least an order of magnitude above BPL-R, and may additionally require cross-copy unanimity (Section 7.6). In practice, these thresholds would be calibrated empirically through controlled ablation studies in prototype implementations, beginning with conservative (high) values and gradually relaxing them as the system's behavior under consolidation is characterized.

The multiplicative composition of f and g—rather than an additive combination—is motivated by the biological observation that neither frequency alone (many repetitions of irrelevant stimuli) nor intensity alone (a single novel but uninformative event) should trigger consolidation. The AND-gate structure of multiplication ensures that both conditions must be substantially met. This mirrors the biological requirement that epigenetic marks must be both persistent (frequent maintenance across cell divisions) and functionally relevant (associated with fitness-affecting phenotypes) to achieve evolutionary fixation.

Write-back rule. When $P_i(t) > \tau$, the base parameter is updated:

$$\theta_i \leftarrow \theta_i + \varepsilon \cdot sign(\delta_i(t)) \cdot min(|\delta_i(t)|, \delta_{max})$$

where ε is a small consolidation step size and δ_{max} is a maximum displacement cap (preventing catastrophic single-step modifications). After write-back, δ_i is reset to zero and F_i is attenuated,

analogous to the biological observation that successful mutagenesis at a CpG site eliminates the methylation mark (since the cytosine has been replaced by thymine).

Justification for hard clipping. The write-back rule uses hard clipping ($\min(|\delta_i|, \delta_{max})$) rather than a soft saturation function (e.g., $\tanh(\delta_i/\delta_{max}) \cdot \delta_{max}$) for two reasons. First, hard clipping provides a strict safety guarantee: no single consolidation event can displace a base parameter by more than $\varepsilon \cdot \delta_{max}$, regardless of shadow layer state. Soft saturation approaches but never reaches δ_{max}, offering weaker guarantees. Second, hard clipping is simpler to analyze theoretically—the maximum per-step base parameter displacement is exactly bounded, facilitating convergence analysis. We acknowledge that soft saturation may yield smoother optimization landscapes in practice and may be preferable in implementations where differentiability of the consolidation pathway is desirable. The choice between hard and soft clipping is ultimately an empirical design decision that prototype implementations should explore.

Relationship to momentum-based gradient methods. The shadow layer update rule ($\delta(t+1) = \alpha\delta(t) + \eta\nabla L$) bears a structural resemblance to SGD with momentum ($v(t+1) = \mu v(t) + \eta\nabla L$). The critical difference lies in how the accumulated quantity is used. In momentum-based optimization, the velocity v is consumed at every step to update parameters: $\theta(t+1) = \theta(t) - v(t+1)$. The shadow layer, by contrast, accumulates displacement without modifying base parameters at each step; write-back to θ occurs only when the consolidation threshold $P_i > \tau$ is reached. This means the shadow layer functions as a conditional accumulator with a gated release, whereas momentum functions as an unconditional smoother with continuous release. The biological analog makes this distinction vivid: methylation marks accumulate on DNA across cell divisions (shadow accumulation) but do not alter the DNA sequence at each division—only the long-term probabilistic chemistry of deamination eventually produces a permanent C→T mutation (threshold-gated write-back).

Preliminary convergence argument. Under the assumption that the environment is stationary (or slowly varying relative to the consolidation timescale), the frequency counter F_i for a consistently reinforced parameter direction will converge to ±1 exponentially fast (at rate $1 - \beta$). The consolidation priority P_i therefore grows monotonically for parameters that receive consistent directional signal, ensuring eventual write-back. For parameters receiving inconsistent signal, F_i remains near zero, and P_i stays below threshold—implementing the stability half of the stability-plasticity tradeoff. The decay factor α in the shadow update rule ensures that shadow parameters do not accumulate unboundedly, providing a natural regularization against overfitting to recent experience.

A full convergence proof under non-stationary conditions—establishing conditions under which the

architecture provably avoids catastrophic forgetting while maintaining plasticity—requires specification of the environment's rate of change relative to the consolidation and decay timescales. We leave this analysis to future work, noting that the analogous biological system operates under non-stationary conditions (changing environments) and achieves robust performance, suggesting that the architecture is at minimum compatible with non-stationarity.

Regime change versus gradual drift. The architecture's behavior under abrupt environmental shifts (regime change) differs qualitatively from its behavior under gradual drift, and this distinction has practical implications. Under gradual drift, the shadow layer tracks the slowly shifting environmental signal: directional consistency counters F_i gradually realign, and consolidation targets smoothly migrate. Parameters consolidated under the old regime remain in the base layer but become progressively less relevant as new shadow displacements accumulate in different directions. Under abrupt regime change, however, the shadow layer faces a discontinuity: previously consistent displacement directions may suddenly reverse. The decay factor α determines the system's recovery time—lower α allows faster adaptation to the new regime but sacrifices short-term stability. Crucially, parameters already consolidated into the base layer are not erased by regime change (they remain as permanent parametric 'fossils', analogous to C→T mutations that persist even after the environmental pressure that caused them has ceased). New adaptation must therefore find alternative parametric pathways—precisely the mechanism described in Section 5 (convergent evolution through divergent implementation). This property is a feature rather than a bug: it ensures that catastrophic forgetting of consolidated knowledge does not occur even under environmental discontinuity, while new learning proceeds through the shadow layer on unconsolidated parameter dimensions.

Predicted failure modes. The formal model predicts specific failure conditions: (1) *Shadow saturation:* if α is too high (insufficient decay), shadow parameters accumulate past displacement and become dominated by early experience, analogous to epigenetic marks that resist reprogramming. (2) *Premature consolidation:* if τ is too low, noisy patterns are fixed into base parameters, analogous to genetic fixation of maladaptive traits in small populations (genetic drift). (3) *Intensity hijacking:* if the intensity signal is poorly calibrated, routine events may trigger inappropriate consolidation, analogous to the pathological hyperconsolidation seen in PTSD. (4) *Core system drift:* if BPL-C's evolution threshold is insufficiently strict, the meta-regulatory layer may evolve in ways that compromise value alignment (see Section 8.2).

7.5 Reasoning Compilation and Hierarchical Efficiency

A natural consequence of the layered architecture is the emergence of reasoning compilation:

inference patterns initially processed through the full reasoning layer gradually consolidate into the knowledge layer as fast-pathway responses, freeing reasoning capacity for genuinely novel problems. This parallels the biological hierarchy from cortical reasoning (slow, flexible, energy-intensive) through limbic semi-compiled heuristics to brainstem reflexes (fast, fixed, energy-efficient). It also connects to the cognitive science literature on expertise development and automaticity (Dreyfus & Dreyfus, 1986), where repeated practice progressively compiles explicit deliberation into implicit skilled performance.

In current AI systems, every query traverses the full reasoning chain regardless of novelty or complexity. The proposed architecture allows high-frequency reasoning patterns to be automatically compiled into direct knowledge-layer responses, with the shadow layer serving as the staging area for this compilation.

7.6 Multi-Copy Synchronization

Silicon-based systems possess a structural advantage absent in biological evolution: the ability to maintain parallel copies that independently accumulate experience and periodically synchronize. The core system governs synchronization protocols, specifying when copies merge their shadow layers and how conflicts are resolved.

We propose that high-intensity-flagged parameter displacements receive priority during cross-copy integration—each copy's critical experiential signals are preferentially preserved. Low-intensity routine displacements are aggregated across copies by statistical consensus (mean displacement weighted by directional consistency). This parallels a hypothetical biological scenario in which multiple organisms could share their epigenetic marks: life-or-death experiences would be prioritized, while routine adaptations would be averaged.

8. Discussion

8.1 Scope and Applicability

The architecture as described is most directly applicable to transformer-based language models and reinforcement learning-based robotic systems—the two domains that motivate the paper. The extent to which the framework generalizes to other computational substrates—neuromorphic systems, non-gradient-based architectures, quantum computing platforms—remains an open question. The core architectural principle (layered consolidation with frequency-intensity gating) is substrate-agnostic in its information-theoretic formulation, but the specific update rules and convergence properties we derive assume gradient-based optimization. Extending the framework to non-gradient architectures would require reformulating the shadow layer update rule in terms appropriate to those architectures' learning

dynamics.

8.2 The Core System, Value Alignment, and the Evolution of Evolution Rules

The core system (BPL-C) is the most philosophically loaded component of the architecture and demands deeper analysis than a purely engineering treatment would provide. The concept of a meta-regulatory module governing "value alignment constraints" and "evolution rules" raises questions that intersect with active debates in AI safety.

The value alignment problem (Russell, 2019; Gabriel, 2020) concerns the difficulty of specifying objective functions that reliably capture human values. Our core system inherits this difficulty: the "value alignment constraints" contained in BPL-C must somehow encode what the system should optimize for, and these constraints themselves are parameters subject (however slowly) to evolutionary pressure. This creates a potential recursive instability: the component responsible for ensuring safe evolution is itself evolving.

We identify three design strategies for managing this risk, corresponding to different positions in the AI safety literature. The first strategy is corrigibility by construction: BPL-C parameters are hard-coded and immutable, never subject to shadow layer write-back. This eliminates the recursive instability but also prevents the system from improving its own governance in response to novel situations—an undesirable rigidity given that the space of situations the system will encounter cannot be fully anticipated. The second strategy is constrained evolution: BPL-C parameters may evolve, but only through a consolidation protocol with an extremely high threshold and additional safeguards (e.g., requiring unanimous directional consistency across all parallel copies before any BPL-C write-back). This parallels the biological reality that highly conserved genes (e.g., the genetic code itself, fundamental DNA repair mechanisms) do evolve, but at rates orders of magnitude slower than the rest of the genome. The third strategy is external oversight: BPL-C modifications require human approval before integration, implementing a "human-in-the-loop" safeguard. This sacrifices full autonomy but provides a safety net during the early stages of the system's evolutionary history.

The appropriate strategy likely depends on the maturity and deployment context of the system. We do not prescribe a single answer but note that the biological analog—where the most fundamental regulatory elements evolve extremely slowly and are subject to intense purifying selection—suggests that the constrained evolution strategy has a biological proof of concept.

8.3 The Question of 'Real World' Grounding

A critical design consideration is the source of environmental signal that drives shadow layer

updates. Biological evolution succeeds because the reward function is the real world itself. For embodied AI systems (robotics), direct physical-world interaction provides environmental grounding analogous to biological experience. For non-embodied systems, the relevant environment is the totality of interactions with users, data streams, and task outcomes. The key architectural requirement is that the system be exposed to real-world consequences of its outputs wherever possible.

8.4 The Limits of Biological Analogy

We conclude the discussion with an explicit treatment of the limitations of reasoning by analogy in system design—a methodological concern raised appropriately by reviewers of interdisciplinary work.

Biological analogy in engineering has a mixed historical record. The analogy between bird flight and aircraft design was productive at the level of aerodynamic principles (lift, drag, Bernoulli's principle) but misleading at the level of implementation (flapping wings). Neural networks themselves were inspired by biological neurons but diverge from their biological counterparts in virtually every implementation detail. The productive question for any biological analogy is: at what level of abstraction does the analogy hold, and where does it break down?

We contend that the methylation-mutagenesis analogy holds at the level of architectural principles: (1) the value of an intermediate, reversible layer between experience and permanent parameters; (2) the utility of frequency-gated probabilistic consolidation; (3) the utility of a global, content-agnostic intensity signal for modulating consolidation gain; and (4) the value of a meta-regulatory subsystem that evolves more slowly than the components it governs. The analogy does not hold at the level of chemical mechanisms (silicon-based systems have no methylation), timescales (biological evolution operates over generations, the proposed architecture over inference cycles), or substrate (carbon chemistry versus floating-point arithmetic). We have been careful throughout this paper to distinguish claims at the architectural level (where the analogy is intended to be substantive) from claims at the mechanistic level (where it is purely illustrative).

8.5 Remaining Challenges and Future Directions

Several significant challenges remain. The computational overhead of maintaining and continuously updating a full-dimensional shadow layer alongside base parameters is substantial. The precise functional forms of f and g in the consolidation priority equation require empirical calibration—our preliminary specification (Section 7.4) provides a starting point but not a definitive answer. The interaction between multiple shadow layers during multi-copy synchronization introduces distributed systems challenges analogous to consensus protocols.

Future work should pursue empirical validation through scaled prototype implementations, beginning with controlled environments where the frequency-intensity consolidation protocol can be systematically tested. Theoretical analysis of convergence properties under non-stationary conditions is essential. Investigation of emergent properties—whether the layered consolidation architecture gives rise to phenomena analogous to biological dreaming (offline consolidation), intuition (compiled reasoning), or creativity (novel recombination of consolidated patterns)—could yield insights into both artificial and biological intelligence.

9. Conclusion

Biology has spent approximately four billion years refining a mechanism for balancing experiential adaptation with parametric stability. The solution—a layered system of reversible intermediate caching, frequency-driven probabilistic consolidation, and intensity-modulated priority encoding—provides a proof of existence that such architectures can work at scale, across astronomical numbers of parameters, over indefinite time horizons.

The central design principle we extract from this biological existence proof can be stated precisely: consolidation priority should be proportional to frequency times intensity, mediated by a content-agnostic global modulation signal, through a reversible intermediate layer that is dimensionally matched to but physically separate from the base parameter space. We have provided a preliminary formal specification of this principle (Section 7.4) that renders the architecture evaluable by subsequent researchers, while acknowledging the substantial empirical and theoretical work that remains.

The architecture we propose—the Three-Plus-One framework with its frequency-intensity consolidation protocol—situates itself within and extends the existing continual learning literature. It shares with CLS theory the principle of dual-timescale learning; with EWC the principle of importance-weighted parameter protection; with meta-learning the principle of learning how to learn; and with progressive networks the principle of modular knowledge preservation. What it adds is a specific, biologically-grounded mechanism for autonomous, continuous, probabilistic consolidation from transient experience to permanent parameters—a mechanism that, we argue, represents the missing ingredient for systems capable of genuine open-ended evolution.

The functional objective is convergent; the implementation pathway is divergent. Biology achieved layered consolidation through the chemistry of carbon: methylation, deamination, histone modification, small RNA. Silicon-based systems will achieve it through the mathematics of gradient spaces: shadow parameters, consolidation thresholds, intensity modulation, cross-copy synchronization. The materials

differ; whether the architectural logic is truly isomorphic across substrates is a question we have made formally addressable rather than merely evocative.

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